Introduction to Databases (CSE 3151)

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B.Tech CSE/CSIT – 6th Semester – 2025–26

IMPORTANT QUESTIONS & EXPECTED END-TERM QUESTION BANK

Based on: PYQ May-2025 | Lesson Plan (Topics 1–43) | All Theory Assignments | DBMS Long Notes

Exam Pattern (End-Semester): Full Marks: 60 | Time: 3 Hours | 10 Questions, each with (a), (b), (c) sub-parts of 2 marks each.
Legend: ★ = Highly Expected (appeared in PYQ or assignments) • = Moderate priority **[PYQ]** = Direct PYQ **[GATE]** = GATE question

Syllabus Units:

- **Unit 1 (CO1):** DBMS Introduction, Data Abstraction, Schema/Instance, Data Independence, DB Languages (DDL/DML), Data Models, DB Architecture (2-tier, 3-tier), Query Processor, Storage Manager
- **Unit 2 (CO3):** ER Model (Entity, Attribute types, Relationship, Degree, Cardinality, Keys), Weak Entity, EER (Specialization, Generalization, Aggregation), ER to Relational Mapping, Schema Diagram
- **Unit 3 (CO4):** Relational DB design, Functional Dependencies, Attribute Closure, Canonical Cover, Decomposition, 1NF, 2NF, 3NF, BCNF, Multivalued Dependency, 4NF
- **Unit 4 (CO5):** Relational Algebra (all operators incl. Division, Outer Joins), Tuple Relational Calculus (TRC), Domain Relational Calculus (DRC)
- **Unit 5 (CO6):** Transactions, ACID Properties, Schedule types, Conflict/View Serializability, Recoverability, Cascadeless schedules, 2PL, Timestamp-based Concurrency Control, Log-based Recovery (REDO/UNDO), Deferred/Immediate Modification

Note: SQL is NOT in the theory syllabus.

UNIT 1 – DBMS Introduction, Architecture & Data Models

(Mapped to Q1 of PYQ – 6 marks per set)

Key Topics: Data Model Types, DBMS Architecture, Languages, Storage & Query Manager

Q1. ★ [PYQ] What is a data model? Differentiate between the **relational** and **semi-structured** data model with examples. *[2 marks]*

Q2. ★ [PYQ] Explain the roles of the **query processor** and the **storage manager** in DBMS architecture. What are the components of the storage manager? *[2 marks]*

Q3. ★ [PYQ] How do **DDL** and **DML** statements affect the **instances** and **schemas** of a database? Give examples. *[2 marks]*

Q4. ★ Explain the three levels of **data abstraction** (Physical, Logical, View) in DBMS with a diagram. What is data independence? *[2 marks]*

Q5. • Distinguish between **physical schema**, **logical schema**, and **view schema**. Define schema and instance. *[2 marks]*

Q6. ★ Differentiate between **two-tier** and **three-tier** database architecture. Where are JDBC/ODBC used? *[2 marks]*

Q7. • Compare DBMS with a traditional **File System** on at least 4 parameters. What are the disadvantages of file-based systems? *[2 marks]*

Q8. • Differentiate between **hierarchical**, **network**, and **relational** data models with examples. *[2 marks]*

Q9. ★ Explain the roles of a **Database Administrator (DBA)**. List at least 5 responsibilities of a DBA. *[2 marks]*

Q10. • What are the different types of database users? Distinguish between **naive users**, **application programmers**, and **sophisticated users**. *[2 marks]*

Q11. ★ Define: (i) Logical data independence, (ii) Physical data independence. How do they differ? *[2 marks]*

Q12. • What is the role of the **buffer manager** and **file manager** in the storage manager component of DBMS? *[2 marks]*

> **Expected Q1 pattern (as in PYQ):** Three sub-parts (a), (b), (c) each 2 marks:
> (a) Data model types / Three-level abstraction / Schema vs Instance
> (b) Query processor & Storage manager roles / 2-tier vs 3-tier architecture
> (c) DDL/DML effect on schema & instance / Data independence types

UNIT 2 – Entity-Relationship Model & ER to Relational Mapping

(Mapped to Q2(a,c), Q3(a,b,c) of PYQ – 10 marks)

2A. ER Model Concepts

Q1. ★ [PYQ] Discuss the **binary** and **ternary** degree of relationships in the ER model with suitable examples. *[2 marks]*

Q2. ★ [PYQ] Distinguish between **attribute inheritance** and **participation inheritance** in specialization with examples. *[2 marks]*

Q3. ★ Explain the different **types of attributes** in the ER model: simple, composite, multi-valued, derived, and key attributes. Give examples for each. *[2 marks]*

Q4. ★ What is a **weak entity set**? How is it represented in an ER diagram? Distinguish between a weak entity and a strong entity. What is an identifying relationship? *[2 marks]*

Q5. • Explain **mapping cardinalities**: one-to-one, one-to-many, many-to-one, many-to-many with examples. How are participation constraints (total/partial) represented in ER diagrams? *[2 marks]*

Q6. ★ Explain **Specialization** and **Generalization** in EER model. What are the constraints: disjoint, overlapping, total, and partial? *[2 marks]*

Q7. • What is **Aggregation** in the ER model? When is it used? Give a suitable example. *[2 marks]*

Q8. ★ Distinguish between **super key**, **candidate key**, **primary key**, **alternate key**, and **foreign key** with examples. *[2 marks]*

2B. ER Diagram Construction Problems

Q1. ★ [PYQ] An entity set **order** with attributes `order_num`, `order_date`, and `payment_info`, and another entity set **order_item** with attributes `item_num`, `quantity`, and `price` are associated with a relationship set "contains". The existence of `order_item` is dependent on the `order`. Construct the suitable ER Diagram. *[2 marks]*

Q2. ★ Design a database for an **online bookstore** with entities: `Book` (ISBN, title, year, price), `Author` (author_id, name, address), `Publisher` (name, address, phone), `Customer` (email, name, address, phone), `Shopping_Basket` (basket_id). Include all relationships with cardinalities and participation constraints. Draw the ER diagram. *[2 marks]*

Q3. ★ Given the following ER diagram of a bank database with entities Customer, Account, Branch, Loan and relationships Depositor, Borrower, Account_Branch, Loan_Branch: (i) Convert to relational schemas, (ii) Identify primary and foreign keys, (iii) Draw the schema diagram. *[2 marks]*

Q4. • Design an ER diagram for a **university** database with entities: Student, Course, Instructor, Department, with appropriate relationships (enrolls, teaches, belongs_to). Show cardinality constraints. *[2 marks]*

2C. ER to Relational Schema Mapping

Q1. ★ **[PYQ]** Convert the given ER diagram (with INSTRUCTOR–advisor–STUDENT relationship and ISA/disjoint specialization on Account into Saving and Checking accounts) to the **minimum number of relational schemas**. Identify primary and foreign keys. *[2 marks]*

Q2. ★ State the rules for mapping the following ER features to relational schemas: (i) Many-to-many relationship, (ii) Weak entity set, (iii) Multi-valued attribute, (iv) Total participation in one-to-one relationship. *[2 marks]*

Q3. • **[PYQ]** Analyze the relational schema resulting from ER-to-relational mapping and **draw the schema diagram** showing primary keys, foreign keys, and referential integrity constraints. *[2 marks]*

Expected Q2–Q3 pattern:
Q2(a) – Binary/ternary relationships OR specialization/generalization concepts
Q2(c) – Attribute/participation inheritance in EER
Q3(a) – Construct ER diagram for given scenario
Q3(b) – Convert ER diagram to relational schema (diagram given)
Q3(c) – Draw schema diagram of the resulting relational schema

UNIT 3 – Normalization & Relational Database Design

(Mapped to Q2(b), Q4(a,b,c), Q5(a,b,c) of PYQ – 14 marks)

3A. Functional Dependencies, Closure, Canonical Cover

Q1. ★ [PYQ] [GATE] Consider a relational schema $R = (E, F, G, H, I, J, K, L, M, N)$ and the set of FDs: $EF \to G$, $F \to IJ$, $EH \to KL$, $K \to M$, $L \to N$ on R. Find the **candidate key** and at least one **super key** of R. *[2 marks]*

Q2. ★ Compute the **closure** of the following set F of functional dependencies for relation schema $r(A, B, C, D, E, F)$: $A \to BC$, $CD \to E$, $B \to D$, $E \to A$. List all candidate keys for r. *[2 marks]*

Q3. ★ [PYQ] Find the **canonical cover** of the following set F of functional dependencies on the schema $R(A, B, C)$: $F = \{A \to BC,\ B \to C,\ A \to B,\ AB \to C\}$. Show all steps. *[2 marks]*

Q4. ★ Consider the set F of FDs on $r(A, B, C, D, E, G)$: $F = \{A \to BCD,\ BC \to DE,\ B \to D,\ D \to A\}$.

(a) Prove using Armstrong's axioms that AG is a super key.
(b) Compute the canonical cover of F (show each step).
(c) Give a 3NF decomposition of R based on the canonical cover.
(d) Find the BCNF decomposition of the schema r.

[8 marks]

Q5. • State and explain **Armstrong's Axioms** (Reflexivity, Augmentation, Transitivity) and their derived rules (Union, Decomposition, Pseudo-transitivity). Prove Transitivity from the basic axioms. *[2 marks]*

Q6. ★ Define **extraneous attribute**. How do you check whether an attribute is extraneous in the LHS or RHS of an FD? Give an example. *[2 marks]*

3B. Normal Forms – 1NF, 2NF, 3NF, BCNF, 4NF

Q1. ★ [PYQ] Explain **First Normal Form (1NF)** with a suitable example. Why is 1NF considered a foundational step in normalization? *[2 marks]*

Q2. ★ [PYQ] What is **partial dependency**? Identify the number of partial dependencies in the relational schema $R(A, B, C, D, E)$ with FD set $F = \{AB \to E,\ E \to C,\ C \to AB\}$. *[2 marks]*

Q3. ★ [PYQ] Consider the relation schema $R(A, B, C, D, E)$ with FD set $F = \{AB \to E,\ E \to C,\ C \to AB\}$. Comment whether it satisfies **2NF** or not with reason. If no, redesign the schema satisfying 2NF. *[2 marks]*

Q4. ★ [PYQ] Consider the relational schema `Student_Performance(Name, CourseNo, RegNo, Grade)` with FDs: $F = \{(Name, CourseNo) \to Grade,\ (RegNo, CourseNo) \to Grade,\ Name \to RegNo,\ RegNo \to Name\}$. Comment whether it satisfies **BCNF** or not with reason. If no, redesign to BCNF. [GATE] *[2 marks]*

Q5. ★ [PYQ] Consider the relation schema $R(A, B, C, D)$ with FD set $F = \{A \twoheadrightarrow D,\ A \to B,\ A \to C\}$. Comment whether it satisfies **4NF** or not with reason. If no, redesign to 4NF. *[2 marks]*

Q6. ★ Consider the relation schema `Student_mark(regd, name, course_id, title, grade)` with FD set $F = \{regd \to name,\ course\_id \to title,\ (regd, course\_id) \to grade\}$. In what normal form is this schema? Check 2NF. If not in 2NF, find the 2NF decomposition and check properties. *[2 marks]*

Q7. ★ Consider the relation schema `Book(Title, Author, Catalog_no, Publisher, Year, Price)` with FDs: $(Title, Author) \to (Catalog\_no, Price)$, $Catalog\_no \to Title$, $Catalog\_no \to Publisher$, $Catalog\_no \to Year$. Comment whether it satisfies **3NF**. If not, find the 3NF decomposition and check properties. *[2 marks]*

Q8. ★ Consider the relation schema $r(PAN, PI, DI, DRUG, QTY, COST)$ with FDs: $PAN \to PI$, $PI \to DI$, $(PI, DRUG) \to QTY$, $(DRUG, QTY) \to COST$.

(a) Check whether it satisfies 3NF. If not, find 3NF decomposition.
(b) Check whether it satisfies BCNF. If not, find BCNF decomposition.

[4 marks]

Q9. ★ A database of journal articles uses the schema: `(VOLUME, NUMBER, STARTPAGE, ENDPAGE, TITLE, YEAR, PRICE)`. Primary key is (VOLUME, NUMBER, STARTPAGE, ENDPAGE). FDs: $(VOL, NUM, SP, EP) \to TITLE$, $(VOL, NUM) \to YEAR$, $(VOL, NUM, SP, EP$
$PRICE$. The DB is redesigned as: Schema1(VOL, NUM, SP, EP, TITLE, PRICE), Schema2(VOL, NUM, YEAR). Which is the **weakest normal form** that the new database satisfies but the old one does not? *[2 marks]*

Q10. ★ Consider the schemas: `books(accessionno, isbn, title, author, publisher)` and `users(userid, name, deptid, deptname)` with FDs: $accessionno \to isbn$, $isbn \to title$, $isbn \to publisher$, $isbn \twoheadrightarrow author$, $userid \to name$, $userid \to deptid$, $deptid \to deptname$.

(a) What is the highest normal form satisfied by each schema?
(b) Normalize both schemas to 4NF.

[4 marks]

3C. Decomposition Properties

Q1. ★ Define **lossless-join decomposition** and **dependency-preserving decomposition**. State the conditions for lossless join. Are these properties always achievable together? *[2 marks]*

Q2. • Explain the concept of **multivalued dependency (MVD)** with an example. State the axioms for MVDs. How does 4NF differ from BCNF? *[2 marks]*

Q3. • Explain the various **anomalies** (insertion, deletion, update) in databases with examples. How does normalization eliminate them? *[2 marks]*

Expected Q4–Q5 pattern (from PYQ):
Q4(a) – 1NF explanation with example
Q4(b) – Canonical cover computation (given FDs on small schema)
Q4(c) – Partial dependency identification / count
Q5(a) – Check 2NF on given schema, redesign if needed
Q5(b) – Check BCNF on given schema (likely Student_Performance type), redesign
Q5(c) – Check 4NF (multivalued dependency), redesign

UNIT 4 – Relational Query Languages (RA, TRC, DRC)

(Mapped to Q6(a,b,c), Q7(a,b,c) of PYQ – 10 marks)
Standard Schema for most queries:
`employee(person-name, street, city)`
`works(person-name, company-name, salary)`
`company(company-name, city)`
`manages(person-name, manager-name)`

4A. Relational Algebra

Q1. ★ [PYQ] Write a **relational algebra** query to find the names, street address, and cities of residence of all employees who work for **TCS** and earn more than **40,000** per annum. *[2 marks]*

Q2. ★ [PYQ] Assume companies may be located in several cities. Write a relational algebra query to find all **company names located in every city where Infosys is located**. (Use division operator.) *[2 marks]*

Q3. ★ [PYQ] Consider the following relation R:

A	B	C	D
α	α	10	7
α	β	15	7
β	β	12	5
β	β	23	10

Write the output of: (i) $\pi_{A,B}(\sigma_{C>10}(R))$ (ii) $\pi_{A,B}(\sigma_{A=B\wedge D>5}(R))$ *[2 marks]*

Q4. ★ Write RA queries on the Employee/Works/Company/Manages schema:

(a) Find names of employees earning more than $60,000 and working in TCS and WIPRO.
(b) Find salary and name of employees working in any company in DELHI.
(c) Find salary of employees living in MUMBAI and working under manager JOHN.

[6 marks]

Q5. ★ [PYQ] Consider two relations $R_1(A, B, D)$ with tuples $(1, 5, 2), (3, 7, 4)$ and $R_2(A, C)$ with tuples $(1, 7), (4, 9)$. Find the **left outer join** and **right outer join** of R_1 and R_2. *[2 marks]*

Q6. ★ Write RA expressions on the schema: Student(SID, Name, City, Age), Course(CID, CName, Dept, Credits), Enroll(SID, CID, Marks), Teacher(TID, TName, Dept, Salary), Teaches(TID, CID):

(a) Find all teachers from CSE department with salary less than 80,000.
(b) Find names of all students with their marks and credits enrolled in any course.
(c) Find courses taught by more than one distinct teacher.

[6 marks]

Q7. ★ Write a RA expression using the **division operator** for: Employee(EId, Name), Brand(BId, BName), Own(EId, BId) – find all EIds who own **all** the brands. *[2 marks]*

4B. Tuple Relational Calculus (TRC)

Q1. ★ [PYQ] Write a **Tuple Relational Calculus (TRC)** query to find the name and company name of the employee who lives in the same city as the company for which they work. *[2 marks]*

Q2. ★ Write TRC expressions for:

(a) Name of employees earning more than 60,000 and working in TCS and WIPRO.
(b) Names of all EIds who own all the brands (using Employee, Brand, Own schema).

[4 marks]

4C. Domain Relational Calculus (DRC)

Q1. ★ [PYQ] Write a **Domain Relational Calculus (DRC)** query to display the name and salary of employees working in a company placed in **Delhi** and employees belonging to **Mumbai**. *[2 marks]*

Q2. ★ Write DRC expressions for:

(a) Salary and name of employees working in any company in Delhi.
(b) EIds who own all brands (Employee, Brand, Own schema).

[4 marks]

Expected Q6–Q7 pattern (from PYQ):
Q6(a) – RA query on employee/company schema (filter + join + project)
Q6(b) – TRC query (same city or similar condition)
Q6(c) – DRC query (city-based filter)
Q7(a) – RA division query (find companies in all cities of X / employees owning all brands)
Q7(b) – Outer joins (left and right outer join on two given relations with tuples)
Q7(c) – Apply RA operators on a given concrete table (compute result)

UNIT 5 – Transactions, Concurrency Control & Recovery

(Mapped to Q8(a,b,c), Q9(a,b,c), Q10(a,b,c) of PYQ – 18 marks)

5A. ACID Properties & Transaction States

Q1. ★ [PYQ] Explain how each of the **ACID properties** (Atomicity, Consistency, Isolation, Durability) ensures reliable transaction processing in DBMS. Give an example for each. *[2 marks]*

Q2. • Explain the different **states of a transaction** (Active, Partially Committed, Committed, Failed, Aborted) with a state diagram. *[2 marks]*

5B. Schedules & Serializability

Q1. ★ [PYQ] Let $R_i(z)$ and $W_i(z)$ denote read and write operations on data item z by transaction T_i. Check whether the given schedule S with three transactions is a **conflict serializable** schedule or not (draw the precedence graph):

$$S : R_1(x)\ R_2(x)\ W_1(x)\ R_3(x)\ W_2(x)$$

[2 marks]

Q2. ★ [PYQ] [GATE] Check whether the given schedule S_1 with four transactions is **conflict serializable** and **recoverable** or not. Justify your answer.

$S_1 : R_2(A)\ W_3(A)\ \text{Commit}(T_3)\ W_1(A)\ \text{Commit}(T_1)\ W_2(B)\ R_2(C)\ \text{Commit}(T_2)\ R_4(A)\ R_4(B)\ \text{Commit}(T_4$

[2 marks]

Q3. ★ Check whether the following schedules are conflict serializable or not:

(a) $S_1 : r_1(x)\ r_1(y)\ r_2(x)\ r_2(y)\ w_2(y)\ w_1(x)$
(b) $S_2 : r_1(x)\ r_2(x)\ r_2(y)\ w_2(y)\ r_1(y)\ w_1(x)$

[4 marks]

Q4. ★ Check whether the given schedule S is **conflict serializable** and **recoverable** or not:

T1	T2	T3	T4
	READ(X)		
		WRITE(X)	
		COMMIT	
WRITE(X)			
COMMIT			
	WRITE(Y)		
	READ(Z)		
	COMMIT		
			READ(X)
			READ(Y)
			COMMIT

[2 marks]

Q5. ★ [PYQ] **Prove or disprove**: Every recoverable schedule is cascadeless. Justify your answer with an example. *[2 marks]*

Q6. ★ Consider a schedule of transactions T1 and T2:

T1	RA		RC		WD	WB	Commit	
T2		RB	WB	RD		WC		Commit

Which of the following schedules is conflict equivalent to the above schedule?
(A) T1: RA RC WD WB Commit; T2: RB WB RD WC Commit
(B) T2: RB WB RD; T1: RA RC WD WB Commit; T2: WC Commit *[2 marks]*

5C. Concurrency Control – 2PL & Timestamp Protocol

Q1. ★ [PYQ] Consider schedule S with transactions T1 and T2. T1 transfers Rs.500 from account A to B, and T2 adds Rs.100 into account A. Prepare a concurrent schedule with the **two-phase locking (2PL) protocol**. *[2 marks]*

Q2. ★ Consider two transactions: **T1**: `read(A); read(B); if A=0 then B:=B+1; write(B).` and **T2**: `read(B); read(A); if B=0 then A:=A+1; write(A).` Add **lock and unlock** instructions so that they observe the two-phase locking protocol. Can the execution result in a **deadlock**? *[2 marks]*

Q3. ★ [PYQ] The transactions T1, T2 with timestamps (TS) 5, 10 respectively want to **write** data item X. The current $RTS(X) = 8$ and $WTS(X) = 7$. Will the write be allowed for both T1 and T2? Justify using the **timestamp ordering protocol**. *[2 marks]*

Q4. ★ [PYQ] Given the following transactions and their timestamps, **simulate the timestamp ordering protocol** and determine whether each operation is allowed or leads to a rollback:

T1: TS = 10	**T2: TS = 20**
Read(X)	
	Read(X)
	Write(X)
Write(X)	

Initial: $RTS(X) = 0, WTS(X) = 0$ *[2 marks]*

Q5. ★ Check whether the given schedule can complete all operations using the **timestamp-based protocol** or not:

T1	**T2**
READ(A)	
	READ(A)
WRITE(A)	
	WRITE(A)

(Assume T1 has a smaller timestamp than T2.) *[2 marks]*

Q6. • Explain the **strict 2PL** and **rigorous 2PL** protocols and how they differ from basic 2PL. What advantage do they provide for recovery? *[2 marks]*

Q7. • What is a **deadlock**? How can it be detected using a wait-for graph? How is it resolved? *[2 marks]*

5D. Database Recovery – Log-based, REDO/UNDO

Q1. ★ [PYQ] Write down two major differences between **immediate database modification** and **deferred database modification**. *[2 marks]*

Q2. ★ [PYQ] What actions will be taken by the **log-based recovery technique** for deferred and immediate database modification, if a failure occurs after the Write(B) operation in the following schedule:

T1	T2	T3
Read(A)		
A=A−500		
Write(A)		
Commit		
	Read(B)	
	B=B+100	
	Write(B)	
	Commit	
		Read(C)

[2 marks]

Q3. ★ Consider the following log sequence:

$$\langle T_1 \text{ start}\rangle\ \langle T_1, A, 100, 150\rangle\ \langle T_2 \text{ start}\rangle\ \langle T_2, B, 200, 250\rangle\ \langle T_1 \text{ commit}\rangle\ \langle \text{crash}\rangle$$

(a) Identify which transactions need UNDO and REDO.
(b) Show the final values of A and B after recovery.

[4 marks]

Q4. ★ Explain the **REDO** and **UNDO** operations in log-based recovery. Under what conditions is each applied? Explain checkpointing. *[2 marks]*

Expected Q8–Q10 pattern (from PYQ):
Q8(a) – ACID properties explanation
Q8(b) – Conflict serializability check (precedence graph on 3–4 transactions)
Q8(c) – Conflict serializable + recoverable check (GATE-type)
Q9(a) – Recoverable vs cascadeless proof/disprove with example
Q9(b) – 2PL for bank transfer / add lock-unlock to given transactions
Q9(c) – Timestamp ordering: check if write allowed given RTS/WTS
Q10(a) – Timestamp ordering protocol simulation (table-based)
Q10(b) – Differences: immediate vs deferred modification
Q10(c) – Log-based recovery: REDO/UNDO with given schedule and crash point

COMPLETE MOCK QUESTION PAPER (Based on PYQ Pattern)

END-SEMESTER EXAMINATION – PRACTICE SET

Introduction to Databases (CSE 3151) | Full Marks: 60 | Time: 3 Hours

Answer ALL questions. Each question carries 6 marks (2 marks each for a, b, c).

Q1. (a) What is a data model? Differentiate between the relational and semi-structured data model. [2]

(b) Explain the roles of the query processor and the storage manager in DBMS architecture. [2]

(c) How do DDL and DML statements affect instances and schemas of a database? [2]

Q2. (a) Discuss the binary and ternary degree of relationships in the ER model with suitable examples. [2]

(b) Consider $R = (E, F, G, H, I, J, K, L, M, N)$ with FDs: $EF \to G$, $F \to IJ$, $EH \to KL$, $K \to M$, $L \to N$. Find the candidate key and at least one super key. [2]

(c) Distinguish between attribute inheritance and participation inheritance in specialization with example. [2]

Q3. (a) An entity set **order** with attributes `order_num`, `order_date`, `payment_info` and entity set **order_item** with attributes `item_num`, `quantity`, `price` are linked by relationship "contains". Existence of order_item depends on order. Construct the ER diagram. [2]

(b) Convert the given ER diagram (Instructor-advisor-Student with Account ISA into Saving/Checking) to minimum number of relational schemas. [2]

(c) Analyze the relational schema from (b) and draw its schema diagram. [2]

Q4. (a) Explain First Normal Form (1NF) with a suitable example. Why is 1NF a foundational step in normalization? [2]

(b) Find the canonical cover of $F = \{A \to BC,\ B \to C,\ A \to B,\ AB \to C\}$ on schema $R(A, B, C)$. [2]

(c) What is partial dependency? Identify the number of partial dependencies in $R(A, B, C, D, E)$ with $F = \{AB \to E,\ E \to C,\ C \to AB\}$. [2]

Q5. (a) Consider $R(A, B, C, D, E)$ with $F = \{AB \to E,\ E \to C,\ C \to AB\}$. Comment whether it satisfies 2NF or not. If no, redesign to 2NF. [2]

(b) Consider `Student_Performance(Name, CourseNo, RegNo, Grade)` with $F = \{(Name, CourseNo) \to Grade,\ (RegNo, CourseNo) \to Grade,\ Name \to RegNo,\ RegNo \to Name\}$. Comment on BCNF. If no, redesign. [2]

(c) Consider $R(A, B, C, D)$ with $F = \{A \twoheadrightarrow D,\ A \to B,\ A \to C\}$. Comment on 4NF. If no, redesign. [2]

Q6. (a) Using employee/works/company schema, write a RA query to find names, street and city of employees who work for TCS and earn more than 40,000. [2]

(b) Write a TRC query to find name and company name of employees who live in the same city as the company they work for. [2]

(c) Write a DRC query to display name and salary of employees in companies placed in Delhi and employees from Mumbai. [2]

Q7. (a) Write a RA query to find all company names located in every city where Infosys is located. [2]

(b) Consider $R_1(A, B, D)$ = {(1,5,2),(3,7,4)} and $R_2(A, C)$ = {(1,7),(4,9)}. Find left and right outer joins. [2]

(c) Given relation R with columns A, B, C, D and sample tuples, write output of $\pi_{A,B}(\sigma_{C>10}(R))$ and $\pi_{A,B}(\sigma_{A=B \wedge D>5}(R))$. [2]

Q8. (a) Explain how each of the ACID properties ensures reliable transaction processing in DBMS. [2]

(b) Check whether $S : R_1(x)\ R_2(x)\ W_1(x)\ R_3(x)\ W_2(x)$ is a conflict serializable schedule or not. [2]

(c) Check whether $S_1 : R_2(A)\ W_3(A)\ \text{Commit}(T_3)\ W_1(A)\ \text{Commit}(T_1)\ W_2(B)\ R_2(C)\ \text{Commit}(T_2)$ is conflict serializable and recoverable or not. [2]

Q9. (a) Prove or disprove: Every recoverable schedule is cascadeless. Justify with an example. [2]

(b) T1 transfers Rs.500 from A to B, T2 adds Rs.100 into A. Prepare concurrent schedule with two-phase locking protocol. [2]

(c) T1 (TS=5) and T2 (TS=10) want to write data item X. $RTS(X) = 8, WTS(X) = 7$. Will write be allowed for T1 and T2? Justify. [2]

Q10. (a) Given T1 (TS=10) and T2 (TS=20): T1: Read(X); T2: Read(X), Write(X); T1: Write(X). Simulate timestamp ordering protocol. Initial: RTS(X)=WTS(X)=0. [2]

(b) Write two major differences between immediate and deferred database modification. [2]

(c) What actions will be taken by log-based recovery for deferred and immediate modification if failure occurs after Write(B) in the given T1–T2–T3 schedule? [2]

QUICK REVISION – Formula Sheet & Key Definitions

Normalization Quick Reference

NF	Condition	Violation Example
1NF	All attributes atomic (no multi-valued, no repeating groups)	Phone = {111, 222}
2NF	1NF + No partial dependency (every non-key attr fully depends on entire PK)	Composite PK (A,B): $A \to C$ is partial
3NF	2NF + No transitive dependency, OR for every FD $X \to Y$: X is superkey OR Y is prime attribute	$A \to B \to C$ where A is key
BCNF	For every FD $X \to Y$: X must be a superkey	Violates 3NF's prime attribute exception
4NF	BCNF + No non-trivial multivalued dependency where LHS is not superkey	$A \twoheadrightarrow B$ when A is not superkey

Canonical Cover Steps

1. Apply **Union rule**: Combine FDs with same LHS.
2. Remove **extraneous attributes** from LHS (check if attribute is redundant using closure).
3. Remove **extraneous attributes** from RHS (check if FD is derivable without it).
4. Remove **redundant FDs** (FDs derivable from remaining set).
5. Repeat until no changes.

Timestamp Ordering Protocol Rules

Read(X) by Ti (TS = ts):
If $ts < WTS(X)$: Reject (rollback Ti) – reading old value
Else: Execute, set $RTS(X) = \max(RTS(X), ts)$

Write(X) by Ti (TS = ts):
If $ts < RTS(X)$: Reject (rollback Ti) – overwriting value needed by newer T
If $ts < WTS(X)$: Reject (Thomas write rule: can skip, or rollback)
Else: Execute, set $WTS(X) = ts$

Conflict Serializability – Precedence Graph

Build directed graph: For each pair of conflicting operations (R-W, W-R, W-W on same data item) between T_i and T_j, draw edge $T_i \to T_j$ if T_i's operation comes first. Schedule is conflict serializable iff the graph is **acyclic**.

Recoverability Rules

- **Recoverable**: If T_j reads a value written by T_i, then T_i must commit before T_j commits.
- **Cascadeless (ACA)**: If T_j reads a value written by T_i, then T_i must commit before T_j reads it.
- **Strict**: No transaction reads or writes a data item until the transaction that last wrote it has committed/aborted.
- Cascadeless $\Rightarrow$ Recoverable (but NOT vice versa).

Log-Based Recovery Summary

Situation at crash	Deferred Modification	Immediate Modification
T committed before crash	REDO	REDO
T not committed at crash	No action (not written)	UNDO

Armstrong's Axioms

- **Reflexivity**: If $\beta \subseteq \alpha$, then $\alpha \to \beta$
- **Augmentation**: If $\alpha \to \beta$, then $\gamma\alpha \to \gamma\beta$
- **Transitivity**: If $\alpha \to \beta$ and $\beta \to \gamma$, then $\alpha \to \gamma$
- **Union**: If $\alpha \to \beta$ and $\alpha \to \gamma$, then $\alpha \to \beta\gamma$
- **Decomposition**: If $\alpha \to \beta\gamma$, then $\alpha \to \beta$ and $\alpha \to \gamma$

Relational Algebra Key Operators

- **Selection** $\sigma_\theta(R)$: Selects tuples satisfying condition θ
- **Projection** $\pi_{A_1,\ldots,A_k}(R)$: Selects columns
- **Natural Join** $R \bowtie S$: Join on common attributes
- **Left Outer Join** $R\,\mathrm{LOJ}\,S$: Keeps all tuples of R, fills nulls for unmatched from S
- **Right Outer Join** $R\,\mathrm{ROJ}\,S$: Keeps all tuples of S
- **Division** $R \div S$: Finds tuples in R that match ALL tuples in S
- **Rename** $\rho_X(R)$: Renames relation to X

Top 5 Most Likely Questions for End-Term 2026 (based on PYQ + assignment pattern):

1. Q1: Data model types + Query processor/Storage manager + DDL/DML effect
2. Q4/Q5: Canonical cover + Partial dependencies + BCNF/2NF check and decompose (with GATE-type FDs)
3. Q6/Q7: RA query on employee/company schema + Outer joins + Timestamp ordering simulation
4. Q8/Q9: Conflict serializability (precedence graph) + Recoverable/cascadeless proof + 2PL protocol
5. Q10: Timestamp ordering protocol simulation + Immediate vs Deferred + Log recovery REDO/UNDO